Assessment 4: Dashboard Visualisation Plan

Interactive Power BI Dashboard for the NAPLAN 2014 Queensland Dataset

MA5830 – Data Visualisation

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1 Dashboard Purpose

1a Target audience (domain situation)

The dashboard is built for education-equity analysts and regional officers within a state education authority, with school principals as a secondary audience benchmarking their own school. These users are numerate professionals — fluent in percentages, averages and rankings — but not statisticians: they read charts to support funding submissions and intervention decisions, not to conduct statistical inference. They are time-poor, consulting the dashboard for minutes rather than hours, so every insight must be legible at a glance and defensible when quoted in a briefing.

1b Assumptions

Visualisation. The audience reads bar, column, line and scatter charts and point maps confidently, but not specialist forms such as box or violin plots, so none appear. To improve accessibility for users with colour-vision deficiencies, the AccessibleTidal theme was adopted and red–green encoding was avoided as the primary categorical distinction. Consumption is assumed to be on a desktop browser, so every page is fixed at 1000 × 800 px with no scrolling.

Data. Users accept ACARA data as authoritative and understand ICSEA as a socio-educational index centred near 1000, together with the National Minimum Standard (NMS). School-level aggregation is appropriate because no student-level data are available in the supplied files. Suppressed results (coded ‘--’, ‘^’, ‘*’) are treated as missing, never as zero, so an unreported school is never displayed as a poor performer.

Time. The data cover a single year (2014), so longitudinal trend analysis is not possible and none is implied. Year 3 to Year 9 is presented as a developmental progression across concurrent cohorts, not a time series — stated in an on-page annotation, since the ribbon chart could otherwise be misread as tracking one cohort over six years.

1c Intent

Dashboard intent: answer “where are the equity gaps in Queensland schooling, and which schools and regions need support?” The desired response is that a user identifies a potentially under-performing cohort, region or sector, isolates the schools involved, and uses the evidence to support further investigation or targeted intervention planning. The visualisation-level intents (task abstractions) are set out in Table 1.

Visualisation	Intent (task abstraction)
KPI cards (6)	Summarise — fixed state-wide baseline (schools, mean score, % at NMS, ICSEA)
Scatter — ICSEA vs mean score	Correlate — reference lines split schools into four equity quadrants
Point map — school locations	Locate — geographic spread of the school network
Bar — % meeting NMS by domain	Compare — which learning domain falls furthest below standard
Combo — score & ICSEA by remoteness	Compare / correlate — examine how mean score and ICSEA vary across remoteness categories
Column w/ drill — sector → school	Explore / drill — from sector aggregate down to an individual school
Bar — gap to state average	Deviate — regions above and below the state mean
Bar — score by governing body	Rank — eight provider organisations
Donut — share of schools by sector	Part-to-whole — composition context
Table — school detail	Look up — a named school's score, rank and gap
Matrix — year level × domain	Compare two categories — locate the weakest cell
Columns — score by domain; % NMS by year	Compare — one dimension at a time
Ribbon — sector rank by year level	Track rank change — compare sector ranking across year levels
Decomposition tree	Decompose / explore — investigate how available school characteristics contribute to variation
Report-page tooltip — School Snapshot	Details on demand — per-school score, % at NMS and a domain mini-bar on hover

Table 1. *Intent (task abstraction) of each visualisation, mapped to Munzner's (2014) analysis actions.*

2 Dashboard Layout

The dashboard uses three navigable report pages, each 1000 × 800 px, plus one hidden report-page tooltip. The structure follows Shneiderman's (1996) visual information-seeking mantra — *overview first, zoom and filter, then details on demand* — mapped one stage per page: Overview (state-wide position), School Explorer (filter and drill to individual schools) and Trends & Analytics (analytical and AI-driven investigation).

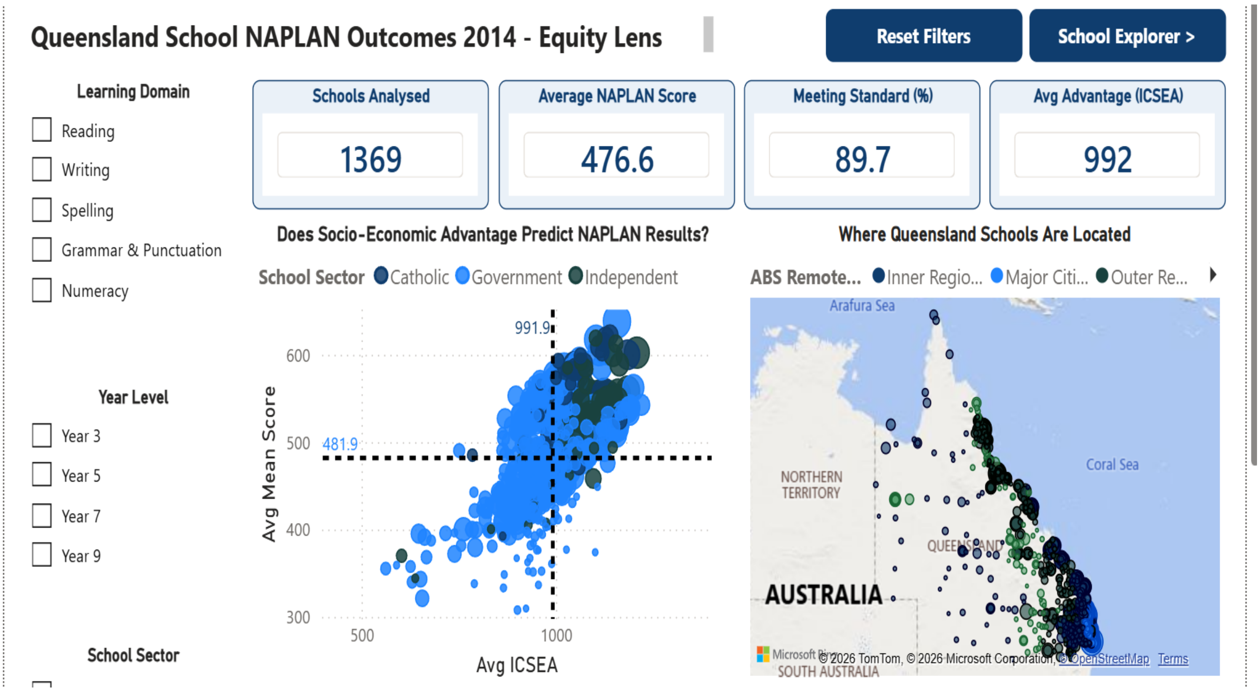

Figure 1. *The Overview page — filter rail (left), KPI card row, the ICSEA-versus-score scatter, and the school location map, with the navigation cluster (top-right).*

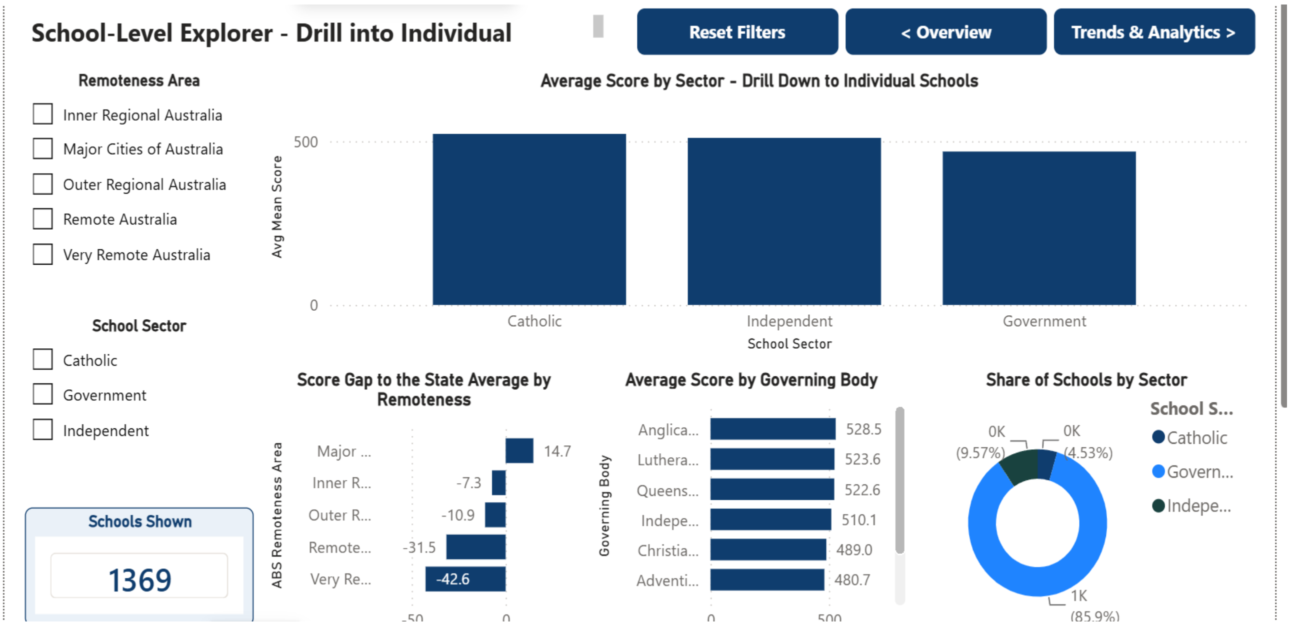

Figure 2. *The School Explorer page — persistent filter rail (left), sector drill-down column chart, supporting charts, and the navigation cluster (top-right).*

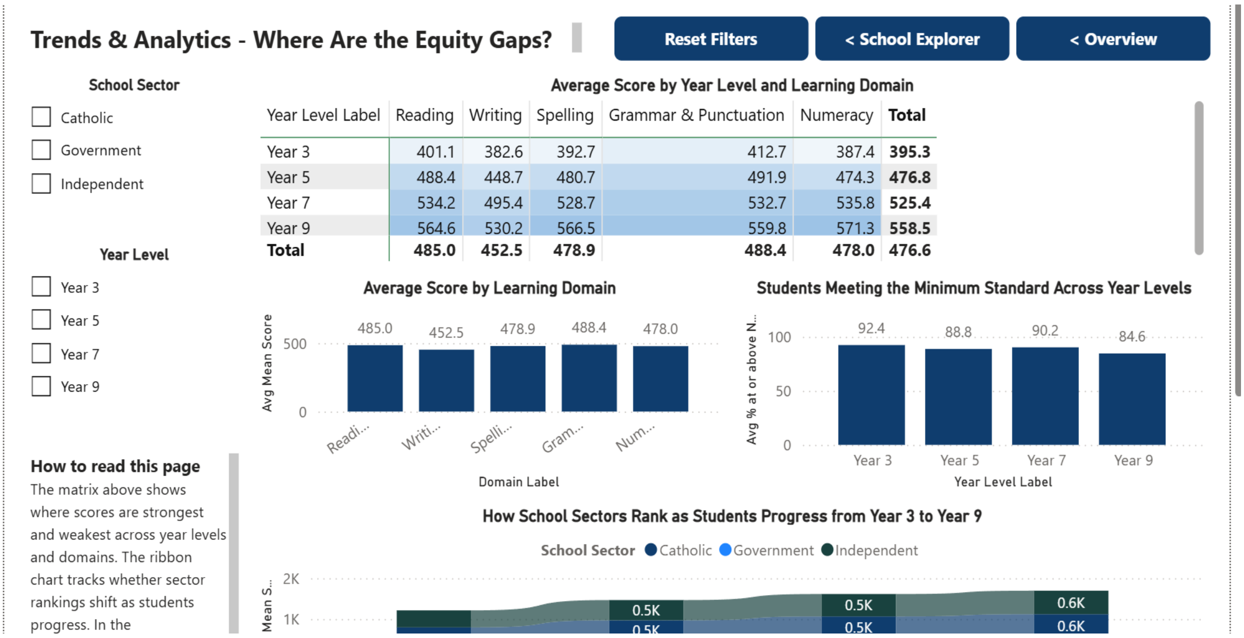

Figure 3. *The Trends & Analytics page — filter rail, the year-level × domain matrix, domain and minimum-standard columns, and the sector ribbon chart.*

Every page repeats this grid, so the eye learns one layout and reuses it — applying the Gestalt principles of proximity and similarity, with consistency serving as an external-cognition aid (Munzner, 2014). Controls are grouped by function: all filters occupy a persistent left rail (two to three slicers per page), and all navigation sits in a right-aligned cluster of identical buttons, top-right. Placing filters on the left and titles at the top-left creates a consistent reading order, while reserving the top-right for actions separates controls from data. Content runs coarse-to-fine down the page: the most aggregated visual first, the row-level table last. This hierarchy also reduces unnecessary visual search because related controls remain in predictable locations across pages. The repeated navigation and filter placement therefore supports comparison between report pages without requiring users to relearn the interface.

3 Dashboard Visualisation Themes

Markers. Marker choice follows each attribute's data type: points for the two-quantitative scatter, bars and columns for categorical magnitude comparison, a line for the secondary ICSEA series in the combo chart, ribbons for rank flow, sized bubbles on the map for spatial volume, and text for KPI values.

Channels. Assignment follows the Cleveland–McGill (1984) accuracy ranking. Quantities that must be judged precisely use position (scatter axes, matrix rows and columns) or length from a common baseline (bar and column charts) — the two most accurately decoded channels. Lower-accuracy channels carry only secondary information: colour hue for unordered categories (sector, remoteness) and area for approximate volume (map bubbles, point size). No quantity that requires precise comparison relies on area alone, because position and length support more accurate quantitative judgements than area (Cleveland & McGill, 1984).

Aesthetics. The dashboard applies the AccessibleTidal theme and uses distinct categorical hues to support visual separation. Chrome is restricted to one navy (#0F3D6E) for cards, buttons and headings, with pale blue (#EAF1F8) for emphasis, so saturated colour appears only where it encodes data. Following Tufte's (2001) data-ink principle, bold weight is rationed to titles, KPI values and annotation headings; axis labels stay at theme default so the marks — not the scaffolding — carry the visual weight.

4 Dashboard Data

All three supplied source files are used. school_stats_naplan_2014_Queensland.xlsx is unpivoted in Power Query from 44 NAPLAN result columns into a 60,236-row fact table; school_locations_2014_Queensland.xlsx, enriched with fields from school_profile_2014_Queensland.xlsx, forms a 1,865-row school dimension. The model links the fact and school tables by ACARA SML ID using a many-to-one relationship with single-direction cross-filtering. This structure keeps school attributes such as sector, remoteness, location and ICSEA in the school dimension while NAPLAN observations remain in the fact table. Consequently, slicers based on school characteristics can filter the NAPLAN measures consistently without duplicating descriptive school attributes across the fact table.

Original and transformed model variables

Original / model variable	**Source · type**	**Used in**
ACARA SML ID	All three source files · integer key	Relationship; school count
Attribute → Value	Derived in Power Query by unpivoting the 44 NAPLAN result columns	Base of all score measures
Year Level (YR3–YR9)	Derived in Power Query from Attribute	Slicers, matrix rows, ribbon axis, legend
Domain (READ/WRIT/SPEL/GP/NMCY)	Derived in Power Query from Attribute	Slicer, matrix columns, bar axis
Measure Type	Derived in Power Query from Attribute	Filter inside every DAX measure
School Name, Suburb	Locations · text	Drill level, detail table, tooltips
School Sector, Governing Body	Locations / Profile · text	Slicers, drill, bar, donut, tree
ABS Remoteness Area, SA4 name	Locations · text	Map legend, combo axis, gap chart
Latitude, Longitude	Locations · decimal	Map position
ICSEA	Profile · integer	Scatter X, combo line, KPI card

Table 2. *Original and transformed model variables carried from the source files into the model.*

Calculated measures, columns and parameters

Derived value	Definition (summarised)	Used in
Year Level Label (col)	"Year " & digit from Year Level	Slicers, matrix rows, ribbon axis
Domain Label / Domain Sort (cols)	SWITCH to readable names; sort column fixes curriculum order	Slicer, axes, matrix columns
Avg Mean Score	AVERAGE of Value on Mean-Score rows	Used throughout the three report pages and the School Snapshot tooltip
Avg % at or above NMS	AVERAGE of Value on Percent-NMS rows	KPI card, comparison charts, ribbon chart, detail table and tooltip
Students Tested	SUM of enrolment rows; REMOVEFILTERS on Domain	Bubble size, point size, tooltips
School Count	DISTINCTCOUNT of ACARA SML ID	KPI cards, donut chart and visual tooltips
Avg ICSEA	AVERAGE of ICSEA	KPI card, scatter plot, remoteness combo chart and tooltips
Mean Score vs QLD Avg	Score − ALL(School_Locations) score	Gap bar, detail table, scatter tooltip
School Rank by Mean Score	RANKX over ALLSELECTED names, ISBLANK-guarded	Detail-table rank column

Table 3. *Derived fields; all measures written in DAX unless marked (col) for a calculated column.*

Data limitations. Three constraints shape interpretation. First, 43.5% of the unpivoted fact-table values are missing after the source suppression/unavailability codes are treated as missing rather than zero; averages therefore use available reported values only. Year 9 has results for 399 schools compared with 904 for Year 3, so cross-year comparisons involve different school populations. Second, ICSEA exists for 1,298 of the 1,369 analysable schools, so the scatter omits 71. Third, 42 school names recur in the dimension, but every name among the 1,369 with results is unique — verified, not assumed — so the ranking measure cannot merge two schools.

5 Dashboard Interactivity

Element	Implementation	Purpose
Filters (slicers)	7 across 3 pages: Domain, Year Level, Sector, Remoteness; multi-select	Narrow the population to a cohort of interest
Filter reset	3 “Reset Filters” buttons (clear all slicers)	Guaranteed return to a known baseline
Cross-filtering	Default across most visuals; 14 deliberate overrides	Selection in one chart re-queries the others
Cross-highlighting	Power BI default retained for chart-to-chart selection (bar, column, donut, ribbon)	Emphasises the selected part while the full distribution stays visible as context
Interaction control	Map & scatter set to none on KPI cards; combo → map and gap → table set to filter	Protect the baseline; direct the focus
Drill down / up	School Sector → School Name hierarchy on the Explorer column chart	Move from aggregate to individual school
Map filtering and navigation	Combo chart set to filter the map; remoteness slicer narrows it; pan and zoom retained	Isolates a region, then inspects dense areas without leaving the page
Navigation buttons	5 labelled buttons, consistent top-right cluster	Movement independent of the page tabs
Tooltips	17 measures across 11 visuals, plus a custom School Snapshot report-page tooltip	Detail on demand without on-canvas clutter
Annotations	3 text panels: Key insight, How to use, How to read	Framing and interaction discoverability

Table 4. *Interactive elements and their purpose across the three pages.*

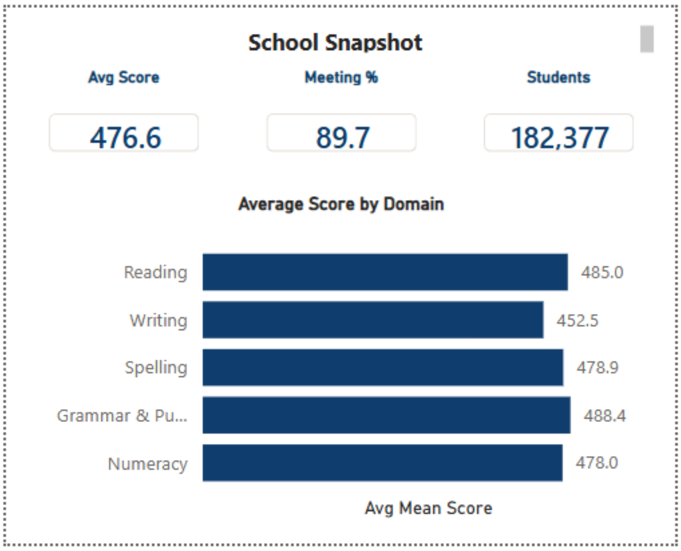

Figure 4. *The custom School Snapshot report-page tooltip, shown on hover — per-school average score, % meeting the standard, students tested, and a domain mini-bar.*

Interaction taxonomy. The elements in Table 4 are not an inventory of Power BI features but deliberate coverage of the interaction intents identified by Yi et al. (2007). Slicers and the reset buttons serve *filter*; selection with highlighting retained serves *select*; drill-down and the report-page tooltip serve *abstract/elaborate*, moving between aggregate and detail without leaving the page; map navigation (pan and zoom) and the page-navigation buttons serve *explore*; and cross-filtering between visuals serves *connect*, exposing how one attribute of a school relates to the rest of its record. Reading the design this way also shows where it is deliberately silent: *encode* and *reconfigure* are withheld, because letting a non-specialist audience change chart types or axes would risk the expressiveness errors that Section 3 sets out to avoid.

Alignment with audience assumptions. Because the audience is time-poor and non-specialist, every page opens unfiltered and immediately readable — no selection is required before the dashboard says something (Few, 2013). Interaction is progressive, not mandatory: slicers and drill-down reward exploration but are never a precondition for insight. The reset buttons provide a reliable return to the baseline view and reduce the risk of interpreting results under unintentionally retained filters. Drill-down is documented in an annotation so that the interaction remains discoverable to users.

Alignment with dashboard intent. The interaction design protects the equity narrative. KPI cards are locked against cross-filtering from the map and scatter so the state-wide baseline stays constant; otherwise clicking one remote school would rewrite the very benchmark against which it is judged. Conversely the combo chart filters the map, so selecting a remoteness band isolates those schools geographically, and the gap chart filters the detail table, so a user who spots a weak region immediately obtains the named schools within it — the desired response defined in Section 1c (Knaflic, 2015). Cross-visual behaviour is therefore governed by a three-tier policy: suppression where a selection would destroy a baseline (twelve interactions set to none), action filtering where a selection should narrow the population (two: combo chart to map, gap chart to detail table), and Power BI's default highlighting everywhere else, which keeps the unselected distribution visible as context rather than discarding it.

Trade-offs acknowledged. Two choices are contestable. A point map was preferred over a choropleth because individual schools, rather than regions, are the unit of intervention; the trade-off is over-plotting in the dense Brisbane corridor, which the remoteness slicer and map zoom only partly mitigate. Locking the KPI cards protects the baseline but breaks the expectation that every visual responds to every selection — a consistency cost accepted deliberately, and the reason an annotation explains what the cards represent.

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